Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/13/2015
PI Name:	Dulchavsky, Scott A. M.D., Ph.D.		
Project Title:	Sonographic Astronaut Vertebral Examination		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads (2) **IVD**:Concern of Intervertebral Disc Damage upon and immediately after re-exposure to Gravity [inactive] (3) **Medical Conditions**:Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures (4) **Renal Stone**:Risk of Renal Stone Formation		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Detroit	**State:**	MI
Zip Code:	48202-2608	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Crew Health NNJ09ZSA002N
Start Date:	06/16/2010	**End Date:**	10/31/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Gilbert, Charlene	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 10/31/2015 per NSSC information and D. Risin/JSC (Ed., 10/13/15) NOTE: End date changed to 4/30/2015 per NSSC information (Ed., 12/16/14) NOTE: Gap changes per IRP Rev E (Ed., 3/19/14) NOTE: Extended to 12/15/2014 per NSSC information (Ed., 3/28/2013)		
Key Personnel Changes/Previous PI:	None		

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Grant/Contract No.:	NNX10AM34G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This proposal responds to NASA Research Announcement (NRA) NNJ09ZSA002N and teams world experts in musculoskeletal (MSK) ultrasound at Henry Ford Health System (HFHS) and space medical experts at Wyle Integrated Science and Engineering to evaluate the ability of ultrasound to characterize acute microgravity associated changes in lumbar and cervical spine to provide operationally relevant data to optimize crew health and guide countermeasure development. An increased height of astronauts is seen during microgravity; the exact mechanism is unclear. Relaxation of the postural muscles, combined with elongation of inter-vertebral disc spaces may be involved. Back pain is frequent during adaption to spaceflight; however, this has not resulted in changes to mission requirements or objectives. Neurologic sequelae have been seen with terrestrial spinal elongation; fortunately this has not occurred in astronaut crews. This proposal will determine the accuracy of MSK ultrasound in characterizing the anatomy of the vertebral unit (disc, musculature, facets, ligaments) and develop just-in-time training methodologies to provide essential information to answer operationally relevant mandates with a team of experts to conduct ground, simulated microgravity, and in-flight experiments to answer the aims: 1. Determine the accuracy of MSK ultrasound in characterizing the normal and microgravity associated changes in the vertebral unit. 2. Investigate the human factors, level of experience, and training necessary to perform focused cervical and lumbar vertebral MSK ultrasound in microgravity with inexperienced ultrasound operators in parallel with expert operators. 3. Determine the anatomic changes in the vertebral unit during long duration spaceflight with astronaut performed vertebral ultrasound. Design and Methods I. GROUND BASED INVESTIGATIONS Baseline evaluation of vertebral unit with MSK ultrasound: Expert sonologists at HFHS will perform magnetic resonance imaging (MRI) and vertebral unit MSK examinations on normal volunteers to assess anatomic accuracy, develop normalization tables, and optimize methodology. Just in time training methodologies for non-expert users A multi-media, just in time training program with reference cue cards will be developed for non-expert operators. The ability of non-expert operators to perform vertebral MSK examinations autonomously or with remote-expert guidance will be evaluated. II. SIMULATED MICROGRAVITY INVESTIGATIONS Human factors analysis of vertebral MSK ultrasound in a microgravity environment will be completed during parabolic flight to optimize subject and operator positioning and restraint. III. FLIGHT INVESTIGATIONS Pre-flight and Post-flight: The flight procedures will consist of a comprehensive vertebral MSK ultrasound and MRI evaluation of long duration crew-members. In-flight procedures will be conducted on the International Space Station (ISS) using the Human Research Facility (HRF) ultrasound at early, mid, and late mission time periods: a post-flight MSK ultrasound and MRI will also be done to assess microgravity anatomic effects and gravitational return to baseline. Significance to NASA: This proposal will provide longitudinal, real time data regarding adaptation of the vertebral unit during long duration spaceflight to enhance mission completion, countermeasure development, and astronaut health. The ability to assess the crewmember musculoskeletal system is critical to guide countermeasures, provide functional data for high risk or impact activities, and assess acute injuries which may occur during exploration class spaceflight. Astronaut performed ultrasound examinations on the ISS have demonstrated that diagnostic quality images can be obtained with targeted training: the procedures developed and verified during this proposal will provide novel data and capabilities to enhance crew health for long duration space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	MRI is the gold standard of spine and intervertebral disk imaging. In many regions of the world there is no access to MRI. Ultrasound is an available and affordable medical imaging option for many areas of the body. However, ultrasound is an unproven imaging modality for the spinal and intervertebral disks (IVD). In this project we are seeking answers to the possibility of using simple off-the-shelf ultrasound to characterize the spine and IVD. The changing spinal dynamics in the microgravity environment allow us an opportunity to prove what is possible for this imaging technique. Additionally, constraints in crew training and the austere environment of the International Space Station allow us to merit this technique in the hands of novice operators using remote guidance as a substitute to intense crew training and just-in-time computer training tools. This research proposal will verify the ability of ultrasound to be used to evaluate changes to the spine of astronauts during spaceflight and patients without access to a CT (computerized tomography) scan or MRI. We are developing new methodologies and training materials to allow non-expert operators to use

	ultrasound to obtain images which would allow distant experts to determine if a significant change or disease process is occurring relative to the cervical or lumbar spine. This information is important in spaceflight; however, spinal disease conditions are widespread on the Earth. This proposal will provide inexpensive, portable diagnostic tools to allow non-physicians to extend medical care capabilities in underserved or geographically isolated areas worldwide.
Task Progress:	We completed a comprehensive analysis of point of care ultrasound for evaluation of the cervical and lumbar spine. Our team demonstrated that expert performed ultrasound was able to rapidly and reproducibly evaluate the cervical spine (C7-C3) and the lumbar spine (S1-L2) for pathology, disc morphology, and inter vertebral spacing. The technique was also expanded to allow the paraspinous musculature to be quantitatively evaluated. We developed a computer based, experiment unique software to provide just in time training in spinal ultrasound for non-expert astronaut ultrasound operators. This software was successfully used, in tandem with brief, hands on training, to teach astronauts and cosmonauts (9 total) the technique of spinal ultrasound to be performed on the ISS. We optimized an MRI protocol to provide high definition images of the lumbar and cervical spine for pre-flight and post-flight analysis against ultrasound. We completed inflight ultrasound evaluations at early, mid, and late mission timepoints using remote expert guidance. All of the evaluations were successfully completed by the crewmembers with excellent image quality that formed the basis of a publication submitted electronically from space (Marshburn TH, Hadfield CA, Sargsyan AE, Garcia K, Ebert D, Dulchavsky SA. "New heights in ultrasound: first report of spinal ultrasound from the International Space Station." J Emerg Med. 2014 Jan;46(1):61-70. http://dx.doi.org/ ; reported in FY2014 Bibliography). Crew preflight, post flight, and in flight imagery was compared to determine the effects of microgravity on the spine. Brief conclusions are summarized; more extensive analysis is ongoing and is shown in the broader final report to NASA. 1. Non-expert operators can perform spinal ultrasound with just in time training and remote guidance. 2. Spinal ultrasound can provide useful morphologic information to evaluate disc height, disc health (dessication), and evaluate pathology (disk herniation, spur formation). 3. Astronaut crewmembers did not have appreciable changes in disc height during spaceflight. 4. Crewmembers did develop isolated pathology (disc alterations, spur formation) during spaceflight. 5. Alterations in the angulation/curvature of the spine occur during spaceflight.
Bibliography Type:	Description: (Last Updated: 07/24/2025)
Articles in Peer-reviewed Journals	Vollman A, Hulen R, Dulchavsky S, Pinchcofsky H, Amponsah D, Jacobsen G, Dulchavsky A, van Holsbeeck M. "Educational benefits of fusing magnetic resonance imaging with sonograms." J Clin Ultrasound. 2014 Jun;42(5):257-63. Epub 2014 Jan 22. http://dx.doi.org/10.1002/jcu.22136 ; PubMed PMID: 24449415 , Jun-2014